

From Design to Use: Understanding Documented Usage of Visual Analytics Systems

Category: Research

Abstract—Understanding how visual analytics systems are designed and used is important for advancing visual analytics theory. Direct records of system use are difficult to obtain at scale, so we turn to published case studies as a source of documented usage. To examine how system design relates to documented use, we organize system functionality, intended workflows, and reported cases into linked, comparable representations. Extracting these representations requires integrating fine-grained evidence distributed across text, figures, and captions while maintaining consistency across the three layers. We develop a skill-guided agentic framework that encodes knowledge specific to visual analytics in reusable instruction packages, or skills, for evidence interpretation, extraction, review, and revision. We apply the framework to 183 visual analytics system papers from IEEE VIS and VAST to construct a knowledge base of documented usage. Human review of 20 papers indicates high extraction quality, with core system and usage semantics well preserved. An ablation study further demonstrates the effectiveness of the skills specific to visual analytics. Corpus-level comparisons and detailed case analysis show that task needs shape how system functionality is used. Views work together to connect findings to further analysis, while intended workflows are realized differently as questions and evidence develop. We also compare language models’ predictions of the next analytic goal and supporting system functionality with and without intended workflows. Providing workflows improves average agreement with documented continuations, although effects vary across cases. Our code, knowledge base, and study materials are publicly available at <https://va-usage.github.io/>.

Index Terms—Visual analytics, usage analysis, knowledge base, data extraction, LLM agent, agent skills

1 INTRODUCTION

Understanding how visual analytics (VA) systems are designed [51] and used [26] is important for advancing VA theory. Prior work on view designs [62], problem-driven design patterns [57], and reusable building blocks [17] has systematically characterized system design. Systematic study of use across systems, however, is constrained by the availability of comparable data [3]. Interaction logs and provenance records [59], as well as field observations [45], depend on access to particular systems and settings, making them difficult to collect and combine at scale. We therefore turn to published case studies, which provide usage accounts across a broad range of VA systems together with their design context. We treat these accounts, mediated by authors’ interpretations and reporting choices [37], as *documented usage*: valuable but selective evidence of system use.

To examine how system design relates to documented use, we bring together three perspectives: what systems provide, how they are intended to be used, and how their use unfolds in reported cases. This approach connects reported actions to the system functions involved and their roles in the intended use of the system. Reliably extracting this interdependent structure requires maintaining completeness and consistency across the three perspectives. The need to integrate textual and visual evidence [27, 53] further complicates extraction from VA papers, where relevant details are scattered across sections, figures, and captions.

We organize these perspectives into linked schema layers for system specification, intended workflow, and documented case. We develop a skill-guided agentic framework to extract the resulting representations from multimodal VA papers. Reusable skills [61] encode knowledge from manual extraction and preliminary failure cases, guiding evidence interpretation, layer-wise extraction, and review and revision. We apply the framework to 183 VA system papers from IEEE VIS and VAST to construct a knowledge base of documented VA usage. Human review of a 20-paper subset indicates high extraction quality, with core system and usage semantics well preserved and most identified issues affecting local details. An ablation on the same subset further demonstrates the effectiveness of the VA-specific skills, with more extraction issues identified without them. We provide *VA-Usage Explorer* for joint inspection of system functionality, intended workflows, and documented cases.

We use the knowledge base to examine how system functionality is used in documented cases and how it helps analysts advance their tasks. Corpus-level statistical comparisons identify patterns and differences

that warrant closer examination, guiding fine-grained analysis of the relevant descriptions in the knowledge base. This analysis addresses three questions: how task needs shape the use of system functionality, how coordination between views supports further analysis, and how task progress shapes the realization of intended workflows in documented cases. We find that the same system functionality can serve different analytical purposes depending on task needs. Coordination between views connects findings to further analysis, while intended workflows are realized differently as questions and evidence develop. Together, the findings suggest that supporting analytical progress requires not only providing relevant functions, but also enabling analysts to adapt how these functions are used and connected as their questions and evidence develop.

Using cases from our knowledge base, we conduct an exploratory comparison of language models’ *next analytic move* predictions with and without the system’s intended workflow. Each prediction identifies the next analytic goal and the system functionality that supports it. Providing workflows improves average agreement with documented continuations, although effects vary substantially across cases. Analysis of illustrative cases suggests that intended workflows can bridge system functionality and its use by clarifying which analytic objective comes next and which operations support it. Our code, knowledge base, interactive explorer, and study materials are publicly available at <https://va-usage.github.io/>.

In summary, we make three contributions:

- A three-layer schema and a knowledge base of 183 VA papers. Analyses of the knowledge base examine how system functionality and intended workflows support documented analytical activity and inform future design.
- A skill-guided agentic extraction framework, evaluated on a human-verified 20-paper subset with a controlled ablation of skill guidance.
- An exploratory next analytic move prediction study examining how intended workflows can connect system functionality to its use.

2 RELATED WORK

We review related work in four areas: organizing knowledge about visual analytics systems, understanding VA usage, knowledge extraction

from research papers, and analytical guidance in VA.

2.1 Organizing Knowledge about Visual Analytics Systems

Prior work organizes knowledge from published visual analytics (VA) systems to support comparison and inform subsequent designs. Domain surveys synthesize this knowledge through shared analytical and design dimensions. Deng et al. [13] relate urban problems to visualization techniques, computational methods, and system organization. Hohman et al. [23] organize deep-learning VA research by analytical goals, users, and usage stages, among other dimensions, while Chen et al. [8] synthesize social-media analysis techniques and pipelines supporting complex tasks.

Complementing these cross-system syntheses, schemas and ontologies explicitly encode selected aspects of system knowledge to support retrieval and reuse. VIS4ML [47] formalizes concepts and relationships in VA-assisted machine-learning workflows as an ontology linked to prior work. VAID [62] indexes composite view designs through descriptions of analytical tasks and visual configurations. Wu et al. [57] connect requirements, data, and solutions to identify problem-driven design patterns, while VA-Blueprint [17] organizes the components and dependencies of urban VA systems into hierarchical building blocks. LEVA [70] uses specifications of visualization designs and view relationships to support LLM-assisted onboarding.

Our knowledge base links system specifications and intended workflows to documented cases, supporting cross-paper analysis of how system functionality and intended processes relate to documented use.

2.2 Understanding Visual Analytics Usage

Research on VA usage draws on observations, interaction records, and published accounts to examine how analytical activity unfolds. Studies with domain experts [26] and field observations of visual investigation [45] provide situated accounts of analytical reasoning. Interaction-log studies [20] and provenance research [40, 59] offer complementary evidence linking recorded activities to sensemaking and insight generation. These approaches provide contextual detail, but collecting comparable usage evidence across systems [3] remains challenging.

Published analysis reports offer a complementary source of usage evidence. Lam et al. [28] derive goal-centered analysis units and their constituent steps from reports in 20 design study papers, establishing a precedent for studying analysis through published narratives. Such reports can communicate the transferable knowledge sought by design study methodology [51], while Meyer and Dykes’s interpretivist account [37] emphasizes that this knowledge remains situated and shaped by researchers’ interpretations.

Our study uses published case studies to examine documented analytical activity, treating these accounts as selective reports rather than complete records of user behavior. This enables literature-scale investigation of recurring analytical patterns and their variation across systems and application domains, providing empirical grounding for reflection on existing designs and informing future VA system design.

2.3 Knowledge Extraction from Research Papers

Prior work addresses two connected problems: recovering structured information from scientific evidence and coordinating the retrieval and verification needed to produce it. Dagdelen et al. [12] demonstrate LLM-based extraction of structured scientific records from text, while the MatViX benchmark [27] extends evaluation to hierarchical outputs grounded in text, tables, and figures.

Tool-using agents combine LLM reasoning with actions such as searching documents and inspecting figures, allowing evidence gathering to adapt to intermediate results. For document understanding, DocAgent [53] coordinates multiple agents to navigate long, multimodal documents. For scientific extraction, nanoMINER [41] coordinates specialized text and image analysis, while STRUCTSENSE [9] combines extraction, ontology alignment, automated evaluation, and human feedback.

General-purpose agents such as Codex [42] and Claude Code [2] offer a complementary route to task specialization. Reusable *skills* [65] package task instructions, reference materials, and scripts to guide an

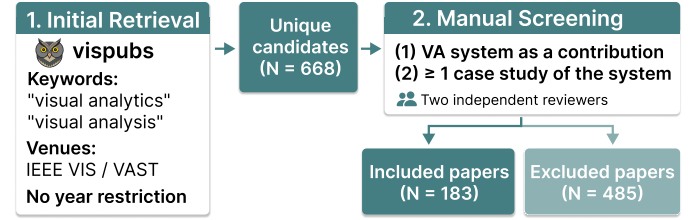


Figure 1: Paper collection pipeline. Keyword- and venue-based retrieval from vispubs yielded 668 unique candidates. Two reviewers independently screened them for both a VA system contribution and at least one case study of the system, confirming 183 included papers through discussion and excluding 485.

existing agent without redesigning its execution architecture. For VA papers, extraction requires interpreting interface figures alongside analytical narratives and maintaining semantic consistency across linked records. We operationalize this approach through VA-specific guidance for evidence interpretation, process decomposition, and cross-layer quality control, organized into extraction, review, and revision and refined through human inspection. Separating these execution procedures from domain-specific guidance offers a reusable workflow pattern for other complex paper-extraction tasks, subject to adaptation and validation of task-specific schemas and skills.

2.4 Analytical Guidance in Visual Analytics

Analytical guidance in VA [7, 10] concerns how systems help analysts determine what to investigate and how to proceed. Prior analytical experience provides one basis for assistance: interaction models [11, 34] use learned patterns to suggest subsequent actions, while Knowledge-Pearls [52] retrieves earlier analysis states from provenance records to support reuse.

LLMs offer a complementary approach by interpreting the current analytical context. LEVA [70] supports onboarding, exploration, and summarization, recommending insights based on the system, analytical task, user interactions, and data. LightVA [69] uses agents to recommend, decompose, and execute analytical tasks, while ProactiveVA [68] infers assistance needs from interaction logs and proactively provides contextual guidance.

We examine whether explicit workflow context helps models predict documented analytic continuations, using linked representations of system functionality, intended workflows, and cases.

3 DATA SOURCE AND SCHEMA

To support literature-scale analysis of documented VA use, we construct a structured knowledge base from VA system papers. This requires a corpus with sufficient usage evidence and a comparable representation of system functionality, intended workflows, and documented cases.

3.1 Paper Collection

We collected papers that present VA systems and document their use through case studies. Figure 1 summarizes the collection pipeline.

Initial Retrieval. We queried the *vispubs* database [29] using the keywords “*visual analytics*” and “*visual analysis*”, restricting venues to *IEEE VIS* and *IEEE VAST* without imposing a publication-year restriction. The search covered records available in the database at the time of retrieval and yielded 668 candidate papers after merging the keyword-query results and removing duplicates.

Eligibility and Screening. We applied two inclusion criteria:

- **System contribution:** The paper explicitly presents a visual analytics system as a contribution.
- **Documented usage:** The paper includes at least one case study of that system.

Two reviewers independently screened all 668 candidates against these criteria and confirmed the final inclusion of 183 papers through discussion.

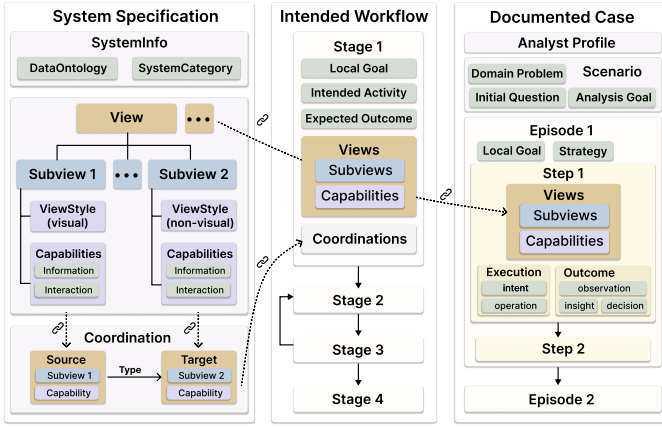


Figure 2: Three linked perspectives on documented VA usage. System specifications describe interface structure and functionality; intended workflows organize expected analysis into stages and directed transitions; documented cases organize reported analysis into episodes and steps. Cross-layer references $\curvearrowright$ connect intended and documented activities to the same system elements, providing a shared basis for design-to-use comparison.

Coverage. The resulting corpus supports cross-paper study of documented usage within this retrieval scope, rather than an exhaustive account of VA system use.

3.2 Knowledge Base Schema Design

Our schema connects three perspectives on design and documented use (Figure 2). System specifications describe what functionality a system provides; intended workflows describe how its authors expect that functionality to be used; documented cases describe how analysis unfolds in specific contexts. The three layers are designed for comparing design and documented use, not as an exhaustive model of analytical practice. Table 1 provides a concise reference for the core terms used throughout the paper; the following paragraphs explain their rationale and scope. Full field definitions and supporting examples are provided in the project resources.

Schema Development. We developed and refined the schema through literature-informed formulation, small-scale manual application, and corpus-scale agentic extraction followed by human review and revision. We formulated an initial schema through coauthor discussions informed by prior literature and the requirements of design-to-use analysis. We then checked its descriptive coverage on five papers spanning different years and domains, revised it through discussion, and examined the revised schema on five additional papers. We subsequently applied the schema at corpus scale through agentic extraction, using the papers and schema definitions as inputs. Our inspection of the resulting representations informed further revisions to the schema and identified recurring extraction difficulties that informed the skill guidance described in the next section.

System Specification. We use LEVA’s specification framework [70] to organize this layer, drawing on other visualization research to refine its components [1, 15] and adding system-level attributes [57] for design-to-use analysis. To accommodate composite visualizations [25], we distinguish a *view*, a top-level logical interface grouping, from a *subview*, a constituent unit with a distinct information or interaction role rather than merely a separate layout region. Drawing on the notion of tool-supported tasks [15], we define a *capability* as what users can do in a subview or what information they can obtain from it. ATWL’s separation of visual representations from derived knowledge [1] motivates capturing information available for interpretation alongside interface operations. An indivisible view has one subview, so capabilities are consistently attached at the same level. Visual subviews describe marks and data-to-channel mappings using a grammar-oriented representation [49], while non-visual subviews describe controls or textual content. Following the linked-interaction principle of coordinated multiple

Table 1: Core schema terms. Stages describe intended analysis; episodes and steps describe reported analysis. Their boundaries reflect analytical purpose rather than individual interface events.

Term	Meaning in this schema
<i>System specification</i>	
View	Top-level logical interface grouping.
Subview	Constituent interface unit with a distinct information or interaction role.
Capability	Information obtainable or actions available through a subview.
Coordination	Directed effect from a source interface unit to one or more targets.
<i>Intended workflow</i>	
Workflow	Author-intended or recommended analytical process, organized into stages and transitions.
Stage	Local phase organized around a dominant analytical goal, with intended activity and supporting system elements.
<i>Documented case</i>	
Case	Coherent reported analysis scenario with its problem, data, analyst context, and overall goal.
Episode	Continuous segment of a case pursuing a dominant local goal; the unit of documented activity compared with workflow stages.
Step	Smallest evidence-supported analytical move within an episode with an independent purpose or result; may comprise several interactions.

views [46], we represent a *coordination* as a directed effect from a source interface unit to one or more targets. At the system level, data ontology annotations characterize dataset structures and attribute properties using the data-abstraction dimensions discussed by Wu et al. [57]. System-category labels identify the reported application domain and task area.

Intended Workflow. This layer represents an author-intended or recommended process of using the system, not a backend data-processing pipeline or a particular case execution. Building on goal-centered analysis units [28] and task-based workflow modeling [16], we define a *stage* as a local phase organized around a dominant analytical goal, with its intended activity, supporting system elements, and expected outcome. Separating purpose, means, and results follows the distinctions in abstract task typologies [4], without adopting a fixed inventory of task labels. Stage boundaries reflect changes in analytical purpose or the role of an activity, not every individual interaction. Following the directed task dependencies in Schulz et al.’s workflow model [50], we define a *transition* as a directed relation indicating an expected move between stages. Our representation allows branching and cycles when supported by the paper, but presentation order alone does not establish an edge.

Documented Case. This layer organizes reported use into cases, episodes, and steps. We use *case* to denote a coherent analysis scenario with its problem, data, analyst context, and overall goal. Following the goal-centered grouping in Lam et al.’s analysis-report framework [28], we define an *episode* as a continuous segment pursuing a dominant local goal. Episodes provide case-specific units for comparison with intended workflow *stages*. A change in the principal subgoal or investigation target establishes a new segment. Building on Gotz and Zhou’s semantic-action abstraction [18], we define a *step* as the smallest evidence-supported analytical move with an independent purpose or result. Several interactions may jointly constitute one analytical move. An episode records its strategy, or overall approach, while steps distinguish immediate intent, performed operations, and documented outcomes. Outcome descriptions distinguish observations from their interpretation as insights, consistent with knowledge-generation models [48], and retain reported decisions separately. Episode and step order follow the reported narrative, without claiming to recover an execution log.

Cross-Layer Connections. Comparing workflow stages with case episodes connects intended analytical processes to their documented realization. We link workflow stages and the constituent steps of case episodes to the same subview and capability identifiers in the system

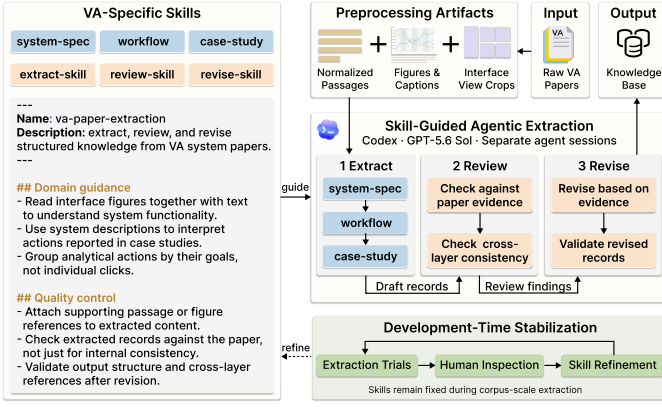


Figure 3: Skill-guided agentic extraction. VA-specific skills, illustrated by representative instructions on the left, guide a general-purpose agent to extract, review, and revise three-layer representations in separate sessions using preprocessed text and images. The revised records populate the knowledge base. Human inspection informs skill refinement during development; the skills remain fixed during corpus-scale extraction.

specification. These shared references provide structural evidence for comparison, while analytical goals and activities provide the semantic context. Sharing a capability does not, by itself, establish an equivalent analytical goal. Stages and episodes retain independently defined boundaries and may differ in granularity or remain unmatched; their correspondence is not restricted to one-to-one matches (Figure 5).

For example, in FluxFlow [67], the thread timeline (*subview*) provides Show Duplicated Users (*capability*), which is referenced by thread drill-down and comparison (*stage*). In the Hurricane Sandy investigation (*case*), cross-thread comparison (*episode*) includes highlighting repeated users in two rumor threads (*step*), using the same capability to identify overlapping participants (*observation*).

4 SKILL-GUIDED KNOWLEDGE BASE CONSTRUCTION

We populate the knowledge base by equipping a general-purpose agent with VA-specific skills for extracting, reviewing, and revising each paper’s three-layer representation. We assess extraction quality through human validation and a skill ablation, and provide an interactive explorer for accessing the resulting knowledge base.

4.1 Framework Overview

The three-layer schema contains fine-grained entities, analytical units, and cross-references whose consistency must be maintained across a paper. Their evidence is inherently multimodal: descriptions of interface structure, functionality, and case-study activities are distributed across text, captions, and figures. We initially used fixed chains of LLM calls, but these proved difficult to adapt to differences in paper structure and evidence distribution. We therefore adopted an agentic approach that can adjust evidence retrieval and task decomposition to each paper.

Agent tools such as Codex and Claude Code [42, 65] combine LLM reasoning with file access and tool execution, allowing them to adapt their actions to intermediate results. *Skills*, reusable packages of task instructions, reference materials, and executable scripts [65], specialize these agents for particular tasks. We build on this separation by encoding VA-specific evidence gathering, interpretation, and quality-control procedures in skills.

Our pipeline (Figure 3) begins with preprocessing, which organizes the paper into identifiable passages, captioned figures, and interface crops, with parsed passages checked against the source PDF. The agent then extracts system specifications, intended workflows, and documented cases in sequence, validating each layer’s structure before proceeding. The system specification supplies shared identifiers for the other two layers, which are constructed separately from their respective evidence. A separate session then reviews the three outputs jointly;

Table 2: Extraction issues identified in 20 papers under the two conditions. With skills: human review; without skills: GPT-6 Astra review against human-corrected gold. Each issue is counted once under its primary layer.

Layer	With skills		Without skills	
	Major	Minor	Major	Minor
System specification	2	27	12	97
Intended workflow	1	0	11	5
Documented case	0	7	13	22
Cross-layer links	0	0	2	10
Total	3	34	38	134

a subsequent revision session addresses supported findings. Our implementation uses Codex with GPT-5.6 at medium reasoning effort for extraction, review, and revision; full prompts, settings, and skill definitions are provided in the project resources.

4.2 Skill Design and Stabilization

Whereas the schema defines what to represent, our skills specify how to gather evidence, decompose extraction tasks, and check outputs.

VA-specific extraction guidance. The guidance addresses three aspects of VA papers: system structure and behavior, analytical activity, and analytical outcomes. First, concepts of visual composition [25] and coordinated views [46] inform how we recover interface structure: images support view decomposition and encoding checks, whereas coordination requires evidence of an interaction’s effect, not merely spatial proximity. Second, goal-centered analysis units [28] and semantic actions [18] guide process decomposition: the skills group activities by analytical purpose rather than individual clicks, while keeping intended processes separate from case-specific narratives. Third, the distinction between findings and their interpretation in knowledge-generation models [48] guides outcome extraction: reported observations must not be expanded into unsupported explanations or decisions. These concepts ground the semantic distinctions; extraction practice informs the corresponding evidence-selection and checking rules.

Execution and quality control. Our skills require source-supported extraction and use the schema’s evidence fields to retain traceable references to passages and figures. Extraction, review, and revision run in separate sessions, allowing review to reassess the outputs against the paper without inheriting the extraction conversation. Review checks evidence support, omissions, analytical boundaries, and cross-layer consistency; revision addresses supported findings and records unresolved uncertainties. Structural and reference validation checks the revised outputs.

Human-guided stabilization. We refined the skills through repeated extraction trials, human inspection of outputs, and coauthor discussion. These trials led us to relax detailed execution instructions in the skills while retaining constraints on evidence and semantics. Human audits helped identify recurring omissions, over-interpretation, and segmentation problems and informed corresponding changes to extraction and review guidance. After stabilization, all papers were processed with the same schema and skill instructions.

4.3 Human Validation

We randomly sampled 20 of the 183 papers while accounting for publication-year distribution. The subset includes 6, 8, and 6 papers from 2013–2017, 2018–2022, and 2023–2026, respectively, covering all 14 years. Four reviewers worked in two pairs, with each reviewer independently reviewing the same 10 papers as their partner. They checked the three extracted layers and their links against the source papers for inaccuracies, omissions, segmentation errors, and invalid links. After completing the independent reviews, the reviewers discussed the identified problems and reached consensus on the issue set.

Using severity criteria derived from the agreed issue set, we classified each independently correctable issue as major or minor and counted it once. *Major* issues change the core meaning of an extracted element

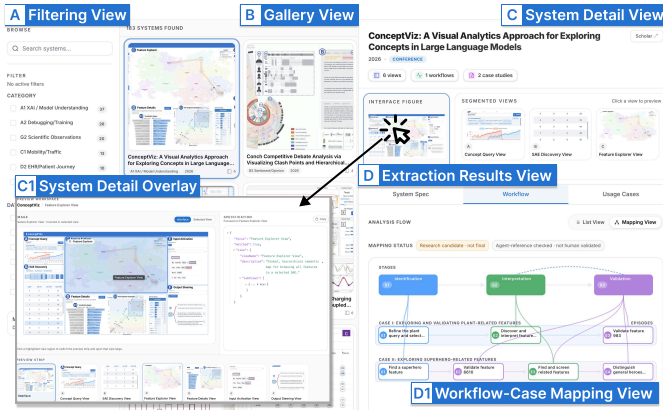


Figure 4: The VA-Usage Explorer interface consists of a filtering view (A), a gallery view (B), a system detail view (C), and an extraction results view (D). The system detail view supports inspection via an overlay panel (C1). The extraction results view (D) presents the three-layer representations and includes a workflow–case mapping subview (D1).

or relationship. Examples include omitting a core function or essential stage, misidentifying an analysis object or comparison baseline, and changing a principal finding or committed action. *Minor* issues, such as a missing auxiliary operation or an imprecise encoding description, affect local details while preserving these core meanings. Across the 20 reviewed papers, we identified 3 major and 34 minor issues (Table 2).

Of the 34 minor issues, 14 concern visual encoding descriptions (marks, channels, or layout) in system specifications, with the represented objects, variables, and functionality remaining correct. These issues may reflect constraints of the representation grammar and difficulties interpreting complex views. Our subsequent analyses do not use these encoding details. Overall, the review suggests that the extraction largely preserves core system and usage semantics, with most identified issues affecting local details.

4.4 Skill Ablation

To assess skill guidance, we evaluated extraction without VA-specific skills on the same 20 papers using the same extraction model, schema, and preprocessed inputs. For each paper, the agent received a generic extraction request in a fresh session and could inspect and revise its outputs autonomously.

We initially attempted to assess the without-skills outputs manually, but the large number of issues made manual review difficult. We therefore constructed a gold reference for each paper to support agent-based assessment, correcting the skill-guided outputs according to the issues confirmed through human review and discussion. Codex with GPT-6 Astra (high reasoning effort) then reviewed each without-skills extraction against gold in an independent session. Reviews followed the same counting and major/minor rules as human validation (Section 4.3), allowing semantically equivalent representations. Human spot checks of the resulting assessments supported their overall reliability.

The review identified 38 major and 134 minor issues, including 44 minor issues concerning visual encoding (Table 2). Errors included omitted workflow branches and overstated case outcomes, which the skills address through guidance on analytical structure and evidence interpretation. System-specification errors also propagated into workflow and case links. These findings suggest that combining VA-specific extraction guidance with review and revision improves the completeness and semantic consistency of the three-layer representation.

4.5 Interactive Access to the Knowledge Base

VA-Usage Explorer (Figure 4) provides interactive access to the knowledge base for joint inspection of system functionality, intended workflows, and documented cases.

Users can filter the collection by system categories and data characteristics (A), browse interface previews and paper metadata in the

gallery (B), and select a paper to inspect its interface and corresponding specifications (C, C1). The extraction results view (D) provides access to the complete three-layer representations. Its workflow–case mapping subview (D1) displays candidate correspondences between workflow **stages** and case **episodes**, using the matching approach described in Section 5.1.

The VA-Usage Explorer is publicly accessible at <https://va-usage.github.io/explorer/>.

5 ANALYZING DESIGN AND DOCUMENTED USE

We examine both *how provided capabilities are incorporated into documented activity* and *how they help analysts advance their tasks*. Published accounts selectively describe use rather than exhaustively record behavior. Our analysis proceeds from statistical comparisons of selected schema attributes to interpretation of the detailed descriptions in the knowledge base. The statistics identify patterns in system structure and cross-layer references; the finer-grained descriptions of tasks, activities, and outcomes help explain how the referenced functionality contributes to analysis. Section 5.1 summarizes the three schema layers and relates documented activity to provided functionality. The following sections examine how task needs shape capability use within shared subviews (Section 5.2), how coordination connects inspection to further analysis (Section 5.3), and how task progress shapes the realization of intended workflows (Section 5.4).

5.1 Structural Patterns of Design and Use

The knowledge base contains 183 papers, 229 intended **workflows**, and 471 documented **cases** comprising 1,189 **episodes** and 2,449 **steps**. We first characterize the three layers using the schema terms in Table 1, then examine their relationships.

System specifications. The specifications record 864 views, containing 1,694 subviews and 4,462 capabilities, together with 1,556 coordinations. These distinguish views and their constituent subviews, the capabilities available within subviews, and connections between them. Subview references describe which subviews enter an account; capability references distinguish the functions used within them.

Intended workflows. Of the 229 workflows, 100 are linear chains, 92 contain directed cycles, and 20 are branching or merging acyclic structures, with the remaining workflows exhibiting other structures. Intended processes thus include both sequential progression and opportunities to revisit stages, rather than prescribing a uniformly linear form of analysis.

Documented cases. The largest recorded categories are expert walk-throughs (148), real-world cases (144), and scenario demonstrations (112). Cases have a median of three episodes and five steps. They are compact accounts from different reporting settings, not uniform interaction logs; case type and reporting granularity therefore matter when comparing their coverage.

Relating activity to provided functionality. Across the corpus, 75.4% of provided subviews are referenced in both intended workflows and documented case steps, compared with 47.2% of capabilities. Across 161 papers with multiple cases, 138 reference more capabilities in their combined cases than in any single case, showing how broad paper-level coverage can accumulate across locally focused accounts.

Even when cases reference the same subviews, their documented use can differ. Among 33 within-paper case pairs referencing identical subview sets, 24 reference different capability sets. This raises the question addressed in Section 5.2: *why do cases draw on different capabilities within the same subviews, and how do those capabilities help advance their tasks?*

Bidirectional coordination. Across 96 papers, 281 pairs of distinct subviews have designed connections in both directions. This recurring design structure motivates the question addressed in Section 5.3: *how are the two directions used in documented cases, and what does each contribute to the ongoing task?*

Comparing intended workflows with documented cases. We match workflow stages and case episodes using their references to system functionality and descriptions of analytical goals and operations. An episode may correspond to several stages considered together,

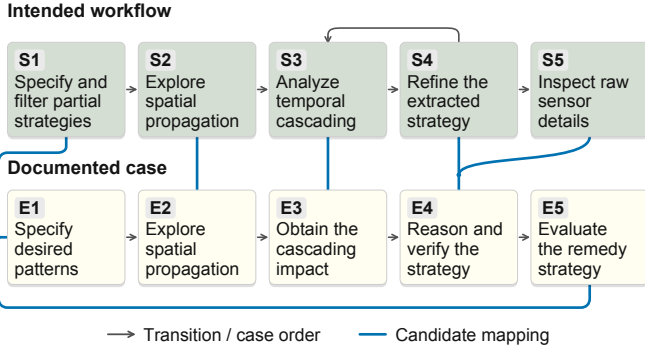


Figure 5: Workflow-case mapping in ECoalVis [35], showing the intended forward-analysis workflow and the documented unbalanced-burning case. Blue lines indicate candidate correspondences; gray arrows indicate workflow transitions or case narrative order. The highlighted relationships have different interpretations. E4 combines strategy refinement (S4) and raw-sensor verification (S5) within one episode, reflecting a difference in segmentation granularity. E1 and E5 both use the strategy-query functionality represented by S1, but for different purposes. E1 selects a strategy to investigate. E5 re-queries the established strategy to assess historical support for the explanation developed during the analysis. Labels are abbreviated from the knowledge base.

and several episodes may jointly correspond to one stage. We score candidate groups of stages and episodes by combining overlap in functionality references with text similarity between their descriptions of analytical goals and activities. The assignment balances these scores against penalties for larger groups and allows episodes to remain unmatched. Figure 5 illustrates how these correspondences can reflect differences in segmentation or the reuse of functionality for different analytical purposes. Under the adopted matching configuration, candidate correspondences cover 96.5% of episodes, and 91.3% of cases are fully matched within at least one workflow. The remainder are left unresolved rather than treated as unrelated.

Subsequent comparisons focus on pairs fully matched to a single shared workflow. Among 320 within-paper pairs of the same known case type, 221 have each case completely matched to exactly one workflow, with that workflow shared by both cases. Other pairs lack a common workflow covering both cases in full or retain multiple workflow candidates. Even within a shared workflow, cases differ in the stages they cover and in how those stages are organized. This motivates the question addressed in Section 5.4: *when cases relate to the same intended workflow, what explains differences in their documented activities, and how do those activities advance their particular tasks?*

5.2 How Task Needs Shape Capability Use

We examine how task needs shape capability use when cases draw on the same subviews. We compare 33 within-paper pairs across 29 papers, retaining cases of the same recorded case type with identical subview coverage: 24 pairs reference different capability sets, while nine reference the same sets. Using the detailed descriptions in the knowledge base, we relate the capabilities involved to case questions, analytical activities, and documented outcomes. The findings connect capability selection to task-specific comparison needs and show that identical capability sets can still serve different analytical purposes.

Task needs shape the construction and revision of comparisons. Within the 24 different-capability pairs, a recurring pattern is that selection, filtering, and ranking establish and revise the evidence needed for a comparison. These functions determine not only which objects are inspected, but also the basis on which they become informative. In one OBTracker case [56], switching between shooting-expectation and passing-probability rankings exposes a discrepancy between the indicators. In the other, analysts adjust weights and select high- and low-performing combinations with the same cutter to construct a contrast for explaining teamwork. Within Team-Scouter [5], a known

replacement need supplies the recruitment target in one case, while squad inspection first reveals a rotation need in another, followed by candidate filtering. The latter account also shows that the highest-ranked external candidate can have a negative predicted contribution relative to the incumbent. Across these cases, shared interfaces support different comparison questions: a difference between indicators, an explanation of performance, or suitability relative to an existing team member.

Comparison conditions can also change as findings emerge, creating a need for a new reference or a more focused comparison. In Action-Evaluator [6], one case examines a model-generated alternative for a weak action, while another returns to player navigation to find a similar, higher-scoring player as a reference. In one ArchExplorer case [63], analysts select architectures under an initial parameter constraint. In the other, they compare neighboring high- and low-performing architectures to check principles derived from inspecting high-performing groups. In both systems, returning to selection supplies evidence for an ongoing question.

A visual sensitivity-analysis case [44] also highlights the importance of clear metric semantics: relative ranking stability was interpreted as a smaller absolute effect. Not all 24 pairs exhibit these patterns. Detailed case descriptions also help distinguish differences in capability use from differences in recorded references.

Identical capability sets can serve different analytical purposes.

Among the nine pairs with identical capability sets, six report different analytical purposes, while three use the capabilities for similar purposes. In EventThread [21], the same clustering adjustment and evolution-reading capabilities help identify patient groups warranting clinical review, compare maintenance practices with subsequent repair patterns, and describe divergence and convergence in academic careers. The shared functions supply evidence for different kinds of judgments, from identifying a group for further investigation to characterizing an association or a temporal pattern. By contrast, both Awesum cases [30] use prompt editing to examine model responses, although they test different aspects of model behavior. Capability references alone therefore do not determine how a function contributes to the task.

Design implication. For systems supporting candidate or reference comparisons, *the conditions that construct a comparison should remain inspectable and revisable alongside its results*. Ranking metrics and weights, eligibility constraints, parameter ranges, and reference objects should be available for revision without losing the question under investigation. Their meanings should be explicit, including whether a metric expresses relative ordering or an absolute effect.

5.3 How Coordination Connects Inspection to Further Analysis

We examine how the two directions of a designed subview connection contribute to ongoing analysis. We select cases in which each direction’s source and target capabilities are referenced together within a documented case step. The references supporting the two directions may come from different steps within the same case. This yields 11 candidate instances across nine papers, each consisting of a pair of directional coordination mechanisms and a documented case. Using the detailed descriptions in the knowledge base, we relate connection descriptions to case questions, operations, and outcomes. The detailed descriptions document how an operation in one subview affects the other in both directions for six papers. For the remaining three, they do not clearly establish this relationship in both directions. Across three of the six papers, inspected objects become conditions for subsequent analysis, while the other three show returns that support evidence checking, modification feedback, or continuity of object identity.

Inspected objects become conditions for subsequent analysis.

The shared pattern is a transition from inspecting an object in one subview to using it to configure further analysis in another. In questionnaire analysis [32], selecting a question combination reveals its response distribution; selecting a respondent category then restricts the population shown in the overview, exposing occupational associations for comparison with the full population. In VATLD [19], a low-confidence group is examined through dimension explanations; a dimension noticed there

is then assigned to the landscape’s horizontal axis, revealing another group for diagnosis. In VITALITY [39], a seed-based query produces candidate papers; a relevant paper rediscovered among the results is added to the seed set before the query is rerun, returning another relevant candidate. The two directions thus play complementary roles: one brings material into inspection, while the other makes what is inspected usable as a population filter, an observation dimension, or a query input.

The other three papers illustrate different uses of the return direction. In ConTour [43], a cluster inspected in pathway context is linked back to its genes and compounds, connecting complementary evidence for an explanation. In the CausalMap case [33], a structural what-if operation is followed by inspection of recomputed sentiment in the overview, making the model’s response available for assessment. In sPortfolio [64], after a time-window change and reclustering, an earlier portfolio identifier helps locate the same object for renewed comparison. These returns support checking, feedback, and continuity without necessarily turning an inspected object into a new analysis condition.

Design implication. *Bidirectional coordination should make clear what is transferred, how it changes the receiving subview, and what analytical action it enables next.* Designers should distinguish population filtering, axis assignment, query-input addition, identity highlighting, and model recomputation, even when they connect the same two subviews. They should also make clear whether setting a condition triggers a computation or requires a separate action, and preserve the objects and conditions needed to interpret the next result.

5.4 How Task Progress Shapes Workflow Realization

We examine how cases develop beyond a shared entry into an intended workflow. Within the comparison pool in Section 5.1, we focus on pairs whose opening episodes map to the same stage set. This yields 19 pairs across 17 papers with strictly nested stage coverage, and 12 pairs across 12 papers with identical coverage but different sequences of matched stage sets. Using the detailed descriptions in the knowledge base for all 31 pairs, we relate these structural differences to case questions, episode goals, and step-level activities and outcomes. The findings show that task-specific evidence and output requirements shape the range of activities, while emerging results can change the objects and conditions of subsequent activity within the same workflow.

Task requirements shape the scope of workflow activities. Our within-paper comparisons identify two recurring purposes for extending analytical activity across eight of the 17 papers in the different-coverage group. The first, observed in four papers, is to supply the references or detail needed for a particular judgment. In Cell2Cell [38], one task focuses on reproducing a contact measurement, while another uses neighboring-contact comparisons and cell-polarization inspection to interpret a contact in its spatial context. Both CloudDet cases [60] include individual-node performance inspection, with one additionally comparing groups of nodes to distinguish characteristic performance patterns. One MolSieve case [22] follows diffusion within a trajectory; another investigates whether a structural change recurs at a different temperature by retrieving and checking similar regions in a second trajectory. An unusual co-ownership relationship in OwnershipTracker [58] prompts tracing dated book records for archival confirmation, extending beyond the network-based quantification of known transfers in the other case. These extensions address the evidence needed for a question, rather than a general need to complete more stages. Additional comparisons alone may not supply all the necessary context: in visual sensitivity analysis [44], some interpretations remain unresolved because city and seasonal information is missing.

The second, observed in the other four papers, is to turn analytical understanding into a testable modification or a usable output. Relationship checking occurs in both Compass cases [14], but the pollution case additionally includes deleting links judged incorrect, turning the assessment into a revised relationship graph. In BOOSTVis [36], the model-improvement case goes beyond explaining training behavior by using subgroup and feature comparisons to propose feature combinations and checking their effects through retraining. The pattern-organization task in MotionFlow [24] builds on the identification and validation activities used in the other task by adding tabs for storing and organizing the

identified patterns. In OrderMonitor [54], one case reports a delayed batch for investigation, while another includes correcting alert labels and reprioritizing the remaining orders to produce a revised handling list. Here, the required output helps explain why activity continues beyond an initial interpretation.

Findings redirect subsequent activity within the same workflow. Our comparisons within the equal-coverage group reveal that, in four of the 12 papers, an earlier result becomes a condition for selecting the next analysis object, making existing workflow activities relevant again. For one player in Action-Evaluator [6], a weak action leads directly to testing an alternative; for another, it first prompts retrieval of a similar, higher-scoring player whose actions provide a reference for the test. In CommonsenseVIS [55], one case uses previously saved examples for model editing, whereas another retrieves further erroneous examples based on an identified failure in movie-related reasoning. Activation results from a verified feature in ConceptViz [31] initiate retrieval of new candidates for semantic screening and input tests, extending the investigation beyond the single-feature focus of the other case. Competitor inspection in PromotionLens [66] provides explanatory context in one case and selects a new discount-simulation target in another, revealing a different predicted strategy preference. Across these comparisons, prior results supply a new selection basis, while the selected objects bring new questions to existing activities.

Design implication. Together, the findings suggest that *workflow design should support continuation according to the evidence or output needed for the current task.* This means providing paths from findings to relevant references or detailed records, from judgments to testable modifications or usable outputs, and from results to new analysis objects. These paths should preserve the originating question, selection conditions, and established findings so that subsequent activities remain connected to the reasoning that motivates them.

6 HOW INTENDED WORKFLOWS INFORM NEXT ANALYTIC MOVE PREDICTION

Building on the analyses in Section 5, we ask whether a VA system’s intended workflow helps models anticipate how an analysis proceeds. We investigate this question through an *exploratory comparison* of next analytic move predictions with and without the system’s intended workflow (Figure 6b).

6.1 Task Formulation

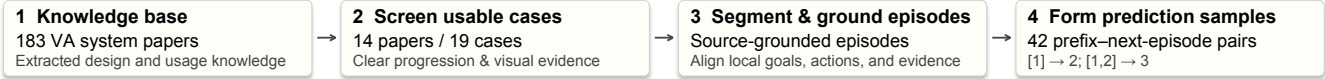
We predict the next local analytic advance from an observed prefix of a published case. The *with-workflow* condition combines four inputs from the knowledge base (Figure 6b). (i) *System functionality* describes available subviews, capabilities, and coordination. (ii) *Intended workflow* supplies all extracted workflows for the system, including their analytic stages, supporting functionality, and transitions. (iii) *Initial case context* states the problem and prior motivation. (iv) *Observed prefix* comprises completed episodes with their goals, steps, and corresponding source images. The *without-workflow* condition omits only component (ii). The workflows describe intended use across cases; neither a current-stage label nor a case-stage alignment is supplied.

The three *Prediction targets* in Figure 6b are the next episode’s *local goal*, *subviews*, and *capabilities*. An episode captures a local analytic advance that may span multiple interactions and visual states; the predictor also gives a short rationale. We evaluate **agreement with a documented continuation**: the next episode reported in the source paper. This reference identifies one realized path, and a different prediction need not be unhelpful in practice. The task does not require predicting a future screenshot or findings from actions not yet performed.

6.2 Prediction Sample Construction

We screen cases in the 183-paper knowledge base using four criteria (Figure 6a, steps 1–2): (1) a reconstructable sequence of analysis with identifiable local goals; (2) intermediate images that align with reported actions and can be assigned to the *Observed prefix* or the *Unobserved next episode* (Figure 6b); (3) prior context supporting interpretation of the continuation without requiring knowledge of an

(a) Construct prediction samples from publications



(b) One prediction sample: ECoalVis / historical remedy analysis

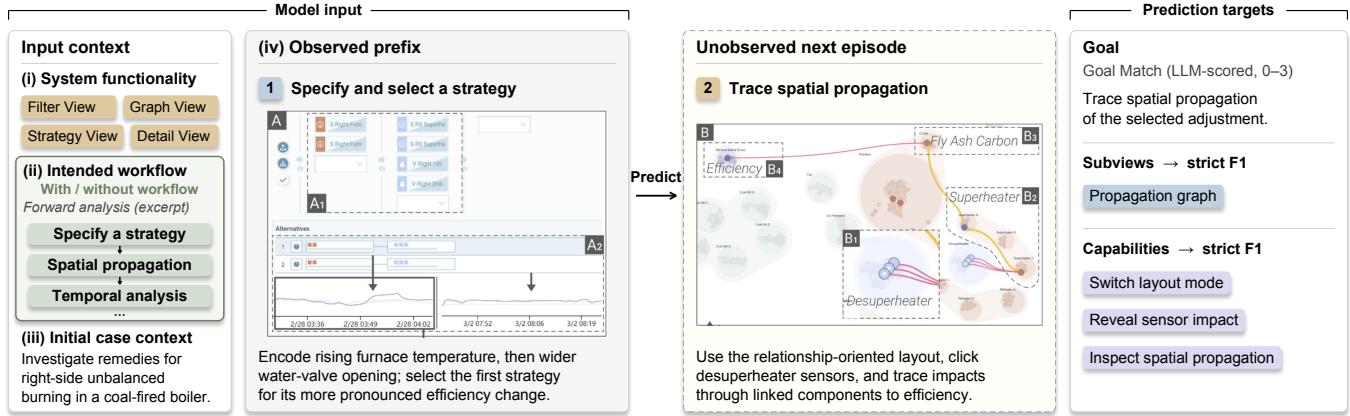


Figure 6: Exploratory comparison with and without a system’s intended workflow. (a) Source-grounded episodes yield 42 samples from 19 cases in 14 papers. (b) In ECoalVis [35], the highlighted workflow (ii) is included or omitted; system functionality (i), initial case context (iii), and the observed prefix with its source image (iv) remain fixed. Episode 1 is observed, and episode 2 supplies the reference targets shown at right. The next episode, its image, and reference labels are withheld from the predictor. Text and tags are abbreviated for display; scoring uses canonical identifiers. Roman numerals identify inputs, numbers identify episodes, and screenshot letters retain source panel labels.

unobserved external event; and (4) source evidence for the next goal and its associated subviews and capabilities. These criteria guide selection from the available documentation; they do not guarantee that the next move is uniquely predictable. The final selection comprises 19 cases from 14 papers.

Step 3 in Figure 6a groups reported actions into episodes by local analytic goal and links each episode to supporting text and image regions. In the ECoalVis example (Figure 6b), specifying a remedy pattern and selecting a historical strategy using source panel A form episode 1; tracing its spatial propagation using source panel B forms episode 2. Investigating the temporal cascade forms a later episode, not shown in the figure. Each cutoff defines a prediction sample (Step 4). Completed episodes form the observed prefix, while the immediately following episode supplies the reference goal, subviews, and capabilities. At the illustrated cutoff, episode 1 is observed and episode 2, including source panel B, is withheld. Multiple cutoffs across the 19 cases yield 42 samples.

The resulting samples retain links to the knowledge base’s case activities, canonical functionality identifiers, and source evidence, supporting inspection of each input and reference continuation.

6.3 Experimental Setup

To examine whether workflow effects recur across predictors, we evaluate three variants from each of two model families: Qwen3.8-Flash, Max, and 27B, and GPT-5.6-Terra, Sol, and Luna. Each model predicts the continuation of all 42 samples under both conditions in three rounds at temperature 0 with reasoning disabled, yielding 1,512 predictions. Repeated generation allows us to inspect the consistency of the comparison.

We measure agreement at three levels. *Goal Match* rates semantic agreement with the reference goal as 0 (no substantive match), 1 (generic relevance), 2 (partial agreement), or 3 (equivalent objective). Qwen3.8-Flash serves as the judge with the same full context for both conditions and no predictor identity. *Subview F1* and *Capability F1* measure exact set agreement on canonical identifiers. To limit the influence of papers contributing more samples, we average prefixes within cases, cases within papers, and then the 14 papers equally.

6.4 Observed Workflow Effects

Table 3 shows that workflow availability increases the mean of both F1 metrics for each of the six models. Averaging the paper-level scores equally across models and rounds, Subview F1 rises from 0.573 to 0.606, Capability F1 from 0.427 to 0.471, and Goal Match from 2.044 to 2.171. The Capability F1 gain mainly comes from better coverage of reference capabilities, with little change in the penalty for predicting capabilities outside the reference. These modest overall increases conceal substantial variation between samples: mean Capability F1 improves in 25, remains unchanged in one, and declines in 16.

Reference episodes vary in scope and may combine several analytic advances. A prediction limited to the immediate next advance can therefore receive low recall against the full reference. The following qualitative analysis examines three cases with improved predictions and interpretable continuations to illustrate possible roles of workflow information (Section 6.5). All 42 samples contribute to the numerical results.

6.5 How Workflow Supports the Next Move

The three cases illustrate complementary roles of workflow information: prioritizing the next analytic stage, specifying supporting operations within that stage, and connecting levels of diagnosis. For each example, F1 is averaged over 18 predictions per condition (six models across three rounds); per-model gains are differences between three-round means.

ECoalVis [35]: organizing the next analytic advance. At the cutoff illustrated in Figure 6b, the user has specified a remedy pattern for unbalanced burning and selected a historical control strategy. In Figure 5, this is the end of E1: the documented continuation E2 traces spatial propagation through the plant (S2), whereas temporal cascading belongs to the later episode E3 (S3). The E2 reference uses one subview and three capabilities, and the corresponding source panel B is withheld from the input. This workflow progression distinguishes the immediate spatial investigation from the later temporal analysis, a distinction that the functionality inventory alone does not express.

The goal predicted by Qwen3.8-Flash in the first round shifts from jointly inspecting spatial propagation and temporal cascading without workflow to tracing spatial propagation across plant components and sensors with workflow. Across all predictions, spatial inspection is

Table 3: Exploratory prediction results on all 42 samples. Without and With indicate workflow availability. Scores are paper-macro means $\pm$ standard deviations across three rounds. Δ is the with-minus-without difference, computed before rounding. Goal Match ranges from 0 to 3; F1 ranges from 0 to 1.

Model	Goal Match			Subview F1			Capability F1		
	Without	With	Δ	Without	With	Δ	Without	With	Δ
Qwen3.8-Flash	2.006 $\pm$ 0.066	2.012 $\pm$ 0.046	+0.006	0.558 $\pm$ 0.020	0.561 $\pm$ 0.019	+0.002	0.384 $\pm$ 0.023	0.415 $\pm$ 0.017	+0.031
Qwen3.8-Max	1.925 $\pm$ 0.085	2.054 $\pm$ 0.110	+0.129	0.562 $\pm$ 0.017	0.573 $\pm$ 0.017	+0.011	0.409 $\pm$ 0.017	0.434 $\pm$ 0.013	+0.025
Qwen3.8-27B	2.010 $\pm$ 0.063	2.198 $\pm$ 0.045	+0.188	0.503 $\pm$ 0.011	0.619 $\pm$ 0.007	+0.116	0.359 $\pm$ 0.009	0.451 $\pm$ 0.012	+0.093
GPT-5.6-Terra	2.038 $\pm$ 0.122	2.196 $\pm$ 0.072	+0.159	0.574 $\pm$ 0.039	0.600 $\pm$ 0.012	+0.026	0.454 $\pm$ 0.020	0.486 $\pm$ 0.010	+0.032
GPT-5.6-Sol	2.190 $\pm$ 0.102	2.248 $\pm$ 0.060	+0.058	0.656 $\pm$ 0.015	0.660 $\pm$ 0.010	+0.004	0.498 $\pm$ 0.009	0.539 $\pm$ 0.010	+0.041
GPT-5.6-Luna	2.093 $\pm$ 0.060	2.315 $\pm$ 0.060	+0.222	0.584 $\pm$ 0.027	0.622 $\pm$ 0.021	+0.037	0.455 $\pm$ 0.016	0.497 $\pm$ 0.023	+0.042

already selected in 15 of 18 without-workflow outputs, but all 18 also include temporal topology and time-lag inspection. With workflow, neither temporal operation is selected; all 18 predictions include spatial inspection and layout switching, and 12 include sensor-impact revelation, the operations illustrated in Figure 6b. Capability F1 rises from 0.272 to 0.933, with gains for every model. The change suggests that workflow helps models narrow the predicted advance to the next stage while identifying its supporting operations.

CommonsenseVIS [55]: specifying supporting operations. After a low-accuracy subset is selected, the workflow connects cluster inspection with concept summaries and subsequent instance analysis. Both conditions identify cluster-alignment inspection in all 18 predictions; workflow increases link hovering from 3 to 14 and cluster selection from 5 to 9. Capability F1 rises from 0.551 to 0.761, with gains for every model. Here, supplying the workflow appears to help models connect an already identified analytic stage to its supporting interactions, making the predicted continuation more complete.

Action-Evaluator [6]: connecting levels of diagnosis. After Mbappé is selected for improvement analysis, the workflow connects diagnosis of match situations to comparison of action choices within a selected situation. With workflow, predictions of sorting action choices rise from 1 to 15 out of 18, and unfolding score details from 0 to 10. Capability F1 rises from 0.366 to 0.616, with gains for every model. The supplied workflow thus appears to help models connect situation-level screening to action-level diagnosis, covering more of the documented continuation.

The selected cases suggest that supplying intended workflows helps models anticipate a documented continuation by clarifying which analytic objective comes next and which operations support it. From a design-to-use perspective, workflow can serve as a bridge between system capabilities and their use by relating available operations to the goals and progression of an analysis.

7 DISCUSSION

Scope of the Knowledge Base. We chose IEEE VIS and VAST to define a focused initial corpus of visualization and VA system research while keeping corpus construction and detailed analysis tractable. Papers from other venues may cover different application areas, system designs, and usage settings. We plan to extend the corpus to CHI and EuroVis papers, as well as TVCG papers beyond the current VIS/VAST scope. This expansion would broaden coverage and allow us to examine whether the observed patterns recur in other publication contexts. We will also continue to track newly published VA system papers and maintain the knowledge base through ongoing updates.

Reliability of Extraction and Analysis. Human validation of 20 papers indicates that the extraction largely preserves core system and usage semantics (Section 4.3). This validation currently covers a subset of the knowledge base. We plan to expand human inspection and correction across the corpus to further strengthen its reliability. We also welcome feedback from system authors and other researchers on the released records. Proposed corrections can be checked against the linked source evidence and incorporated into subsequent knowledge base updates.

Workflow–case matching assumes that documented case activities

can often be related to stages of an intended workflow, although the algorithm may imperfectly capture these relationships. We use the resulting patterns to identify candidate comparisons and examine detailed case descriptions before developing interpretations. This examination allows unsupported correspondences and apparent structural differences to be revised or rejected. Matching limitations may restrict the coverage of the comparison groups, while the reported qualitative findings are supported by the detailed case evidence.

Structural similarities across papers do not by themselves establish comparable analytical roles. We therefore use statistical patterns to guide case selection, interpret each case in its system and task context, and then synthesize commonalities across papers.

Limitations of the Exploratory Prediction Study. On the input side, the main limitation is that the reconstructed context may not fully reflect actual system use. The predictor receives system functionality descriptions, but the current state is represented through selected views and published narratives. In real-world use, users can access other views, adjust selections, and observe interaction feedback that these materials may not capture. Published figures may also span more than one analytical moment, making some episode boundaries less clear. The observed workflow effects should therefore be interpreted within this available context.

On the output side, a VA system may support multiple reasonable analysis paths, while each documented case records a particular continuation. A prediction that differs from this reference may still be appropriate. Episode granularity also affects scoring: a coherent next advance can receive lower recall when the reference combines several advances. The scores thus measure agreement with documented continuations and provide an exploratory comparison of workflow conditions. Broader evaluation with more complete system states, multiple acceptable continuations, and actual user tasks would help assess the value of intended workflows for guidance in practice.

Limitations of Documented Use. Published case studies convey authors’ selected and interpreted accounts of use. They may compress unsuccessful attempts, revisions, and exploratory detours, and provide limited evidence of sustained use, collaboration, or activity outside the system. Our analysis consequently characterizes reported activity in its design context, without establishing everyday usage frequencies or the effectiveness of the reported designs. Studying actual use through observation and interaction records remains important for understanding these aspects. We plan to explore reconstructing selected VA systems using the knowledge base as a starting specification, supplementing it with source materials and checking key functionality and interaction behavior. User studies in these executable environments could capture system states and interaction trajectories together with task context, enabling comparison with documented accounts. The resulting records would describe actual user activity in reconstructed systems and complement the evidence available from publications. These studies could also test the design implications in Section 5. Of particular interest is the context users need when revising comparison conditions, transferring objects between views, or revisiting earlier activities. We could examine whether and how keeping the original question, selection conditions, and established findings accessible helps users continue their analysis.

REFERENCES

- [1] N. Andrienko, G. Andrienko, J. Bernard, and M. Sedlmair. ATWL: A formal language for representing, comparing, and reusing visual analytics workflows, 2026. arXiv:2605.25489. doi: 10.48550/arXiv.2605.25489 3
- [2] Anthropic. How Claude Code works. Official documentation. Accessed September 5, 2026. 2
- [3] L. Battle, M. Angelini, C. Binnig, T. Catarci, P. Eichmann, J.-D. Fekete et al. Evaluating visual data analysis systems: A discussion report. In *Proceedings of the Workshop on Human-In-the-Loop Data Analytics*, pp. 1–6, 2018. 1, 2
- [4] M. Brehmer and T. Munzner. A multi-level typology of abstract visualization tasks. *IEEE transactions on visualization and computer graphics*, 19(12):2376–2385, 2013. 3
- [5] A. Cao, X. Xie, R. Zhang, Y. Tian, M. Fan, H. Zhang et al. Team-Scouter: Simulative visual analytics of soccer player scouting. *IEEE Transactions on Visualization and Computer Graphics*, 31(1), 2025. doi: 10.1109/TVCG.2024.3456216 6
- [6] A. Cao, X. Xie, M. Zhou, H. Zhang, M. Xu, and Y. Wu. Action-Evaluator: A visualization approach for player action evaluation in soccer. *IEEE Transactions on Visualization and Computer Graphics*, 30(1), 2024. doi: 10.1109/TVCG.2023.3326524 6, 7, 9
- [7] D. Ceneda, T. Gschwandtner, T. May, S. Miksch, H.-J. Schulz, M. Streit et al. Characterizing guidance in visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, 23(1):111–120, 2017. doi: 10.1109/TVCG.2016.2598468 2
- [8] S. Chen, L. Lin, and X. Yuan. Social media visual analytics. In *Computer Graphics Forum*, vol. 36, pp. 563–587. Wiley Online Library, 2017. 2
- [9] T. R. Chhetri, Y. Chen, P. Trivedi, D. Jarecka, S. Haobsh, P. Ray et al. Structsense: A task-agnostic agentic framework for structured information extraction with human-in-the-loop evaluation and benchmarking. *arXiv preprint arXiv:2507.03674*, 2025. 2
- [10] C. Collins, N. Andrienko, T. Schreck, J. Yang, J. Choo, U. Engelke et al. Guidance in the human-machine analytics process. *Visual Informatics*, 2(3):166–180, 2018. 2
- [11] F. Dabek and J. J. Caban. A grammar-based approach for modeling user interactions and generating suggestions during the data exploration process. *IEEE Transactions on Visualization and Computer Graphics*, 23(1):41–50, 2017. doi: 10.1109/TVCG.2016.2598471 2
- [12] J. Dagdelen, A. Dunn, S. Lee, N. Walker, A. S. Rosen, G. Ceder et al. Structured information extraction from scientific text with large language models. *Nature communications*, 15(1):1418, 2024. 2
- [13] Z. Deng, D. Weng, S. Liu, Y. Tian, M. Xu, and Y. Wu. A survey of urban visual analytics: Advances and future directions. *Computational Visual Media*, 9(1):3–39, 2023. doi: 10.1007/s41095-022-0275-7 2
- [14] Z. Deng, D. Weng, X. Xie, J. Bao, Y. Zheng, M. Xu et al. Compass: Towards better causal analysis of urban time series. *IEEE Transactions on Visualization and Computer Graphics*, 28(1), 2022. doi: 10.1109/TVCG.2021.3114875 7
- [15] D. Edge, N. Henry Riche, J. Larson, and C. White. Beyond tasks: An activity typology for visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, 24(1):267–277, 2018. doi: 10.1109/TVCG.2017.2745180 3
- [16] P. Federico, A. Amor-Amorós, and S. Miksch. A nested workflow model for visual analytics design and validation. In *Proceedings of the Sixth Workshop on Beyond Time and Errors on Novel Evaluation Methods for Visualization*, pp. 104–111, 2016. 3
- [17] L. Ferreira, G. Moreira, and F. Miranda. Va-blueprint: Uncovering building blocks for visual analytics system design. *IEEE Transactions on Visualization and Computer Graphics*, 2025. 1, 2
- [18] D. Gotz and M. X. Zhou. Characterizing users’ visual analytic activity for insight provenance. *Information Visualization*, 8(1):42–55, 2009. 3, 4
- [19] L. Gou, L. Zou, N. Li, M. Hofmann, A. K. Shekar, A. Wendt et al. VATLD: A visual analytics system to assess, understand and improve traffic light detection. *IEEE Transactions on Visualization and Computer Graphics*, 27(2), 2021. doi: 10.1109/TVCG.2020.3030350 6
- [20] H. Guo, S. R. Gomez, C. Ziemkiewicz, and D. H. Laidlaw. A case study using visualization interaction logs and insight metrics to understand how analysts arrive at insights. *IEEE transactions on visualization and computer graphics*, 22(1):51–60, 2015. 2
- [21] S. Guo, K. Xu, R. Zhao, D. Gotz, H. Zha, and N. Cao. EventThread: Visual summarization and stage analysis of event sequence data. *IEEE Transactions on Visualization and Computer Graphics*, 24(1):56–65, 2018. doi: 10.1109/TVCG.2017.2745320 6
- [22] R. Hnatyshyn, J. Zhao, D. Perez, J. Ahrens, and R. Maciejewski. MolSieve: A progressive visual analytics system for molecular dynamics simulations. *IEEE Transactions on Visualization and Computer Graphics*, 30(1), 2024. doi: 10.1109/TVCG.2023.3326584 7
- [23] F. Hohman, M. Kahng, R. Pienta, and D. H. Chau. Visual analytics in deep learning: An interrogative survey for the next frontiers. *IEEE Transactions on Visualization and Computer Graphics*, 25(8):2674–2693, 2019. doi: 10.1109/TVCG.2018.2843369 2
- [24] S. Jang, N. Elmqvist, and K. Ramani. MotionFlow: Visual abstraction and aggregation of sequential patterns in human motion tracking data. *IEEE Transactions on Visualization and Computer Graphics*, 22(1), 2016. doi: 10.1109/TVCG.2015.2468292 7
- [25] W. Javed and N. Elmqvist. Exploring the design space of composite visualization. In *2012 IEEE Pacific Visualization Symposium*, pp. 1–8. IEEE, 2012. doi: 10.1109/PacificVis.2012.6183556 3, 4
- [26] Y.-a. Kang and J. Stasko. Examining the use of a visual analytics system for sensemaking tasks: Case studies with domain experts. *IEEE Transactions on Visualization and Computer Graphics*, 18(12):2869–2878, 2012. 1, 2
- [27] G. Khalighinejad, S. Scott, O. Liu, K. L. Anderson, R. Stureborg, A. Tyagi et al. Matvix: Multimodal information extraction from visually rich articles. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3636–3655, 2025. 1, 2
- [28] H. Lam, M. Tory, and T. Munzner. Bridging from goals to tasks with design study analysis reports. *IEEE Transactions on Visualization and Computer Graphics*, 24(1):435–445, 2018. doi: 10.1109/TVCG.2017.2744319 2, 3, 4
- [29] D. Lange. Vispubs.com: A visualization publications repository, 2024. doi: 10.31219/osf.io/dg3p2 2
- [30] S. Y.-T. Lee, A. Bahukhandi, D. Liu, and K.-L. Ma. Towards dataset-scale and feature-oriented evaluation of text summarization in large language model prompts. *IEEE Transactions on Visualization and Computer Graphics*, 31(1), 2025. doi: 10.1109/TVCG.2024.3456398 6
- [31] H. Li, Z. Wen, Q. Jiang, C. Li, Y. Wu, Y. Yang et al. ConceptViz: A visual analytics approach for exploring concepts in large language models. *IEEE Transactions on Visualization and Computer Graphics*, 32(1):57–67, 2026. doi: 10.1109/TVCG.2025.3634806 7
- [32] R. Li, W. Cui, T. Song, X. Xie, R. Ding, Y. Wang et al. Causality-based visual analysis of questionnaire responses. *IEEE Transactions on Visualization and Computer Graphics*, 30(1), 2024. doi: 10.1109/TVCG.2023.3327376 6
- [33] R. Li, S. Ye, Y. Lin, B. Zhou, Z. Kang, T.-Q. Peng et al. Causality-based visual analytics of sentiment contagion in social media topics. *IEEE Transactions on Visualization and Computer Graphics*, 32(1), 2026. doi: 10.1109/TVCG.2025.3633839 7
- [34] Y. Li, Y. Qi, Y. Shi, Q. Chen, N. Cao, and S. Chen. Diverse interaction recommendation for public users exploring multi-view visualization using deep learning. *IEEE Transactions on Visualization and Computer Graphics*, 29(1):95–105, 2023. doi: 10.1109/TVCG.2022.3209461 2
- [35] S. Liu, D. Weng, Y. Tian, Z. Deng, H. Xu, X. Zhu et al. ECoalVis: Visual analysis of control strategies in coal-fired power plants. *IEEE Transactions on Visualization and Computer Graphics*, 29(1):1091–1101, 2023. doi: 10.1109/TVCG.2022.3209430 6, 8
- [36] S. Liu, J. Xiao, J. Liu, X. Wang, J. Wu, and J. Zhu. Visual diagnosis of tree boosting methods. *IEEE Transactions on Visualization and Computer Graphics*, 24(1), 2018. doi: 10.1109/TVCG.2017.2744378 7
- [37] M. Meyer and J. Dykes. Criteria for rigor in visualization design study. *IEEE Transactions on Visualization and Computer Graphics*, 26(1):87–97, 2020. doi: 10.1109/TVCG.2019.2934539 1, 2
- [38] E. Mörrth, K. Sidak, Z. Maliga, N. Möller, N. Gehlenborg, P. Sorger et al. Cell2Cell: Explorative cell interaction analysis in multi-volumetric tissue data. *IEEE Transactions on Visualization and Computer Graphics*, 31(1):569–579, 2025. doi: 10.1109/TVCG.2024.3456406 7
- [39] A. Narechania, A. Karduni, R. Wesslen, and E. Wall. VITALITY: Promoting serendipitous discovery of academic literature with transformers & visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, 28(1), 2022. doi: 10.1109/TVCG.2021.3114820 7
- [40] P. H. Nguyen, K. Xu, A. Wheat, B. W. Wong, S. Attfield, and B. Fields. Sensepath: Understanding the sensemaking process through analytic provenance. *IEEE transactions on visualization and computer graphics*, 22(1):41–50, 2015. 2

- [41] R. Odobesku, K. Romanova, S. Mirzaeva, O. Zagorulko, R. Sim, R. Khakimullin et al. Agent-based multimodal information extraction for nanomaterials. *npj Computational Materials*, 11(1):194, 2025. 2
- [42] OpenAI. Codex CLI. Official documentation. Accessed September 5, 2026. 2, 4
- [43] C. Partl, A. Lex, M. Streit, H. Strobelt, A.-M. Wassermann, H. Pfister et al. ConTour: Data-driven exploration of multi-relational datasets for drug discovery. *IEEE Transactions on Visualization and Computer Graphics*, 20(12), 2014. doi: 10.1109/TVCG.2014.2346752 7
- [44] N. Piccolotto, M. Bögl, C. Muehlmann, K. Nordhausen, P. Filzmoser, J. Schmidt et al. Data type agnostic visual sensitivity analysis. *IEEE Transactions on Visualization and Computer Graphics*, 30(1), 2024. doi: 10.1109/TVCG.2023.3327203 6, 7
- [45] I. Rae, F. Zhou, M. Bilsing, and P. Bunge. Understanding visual investigation patterns through digital “field” observations. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*, pp. 1–16, 2022. 1, 2
- [46] J. C. Roberts. State of the art: Coordinated & multiple views in exploratory visualization. In *Fifth International Conference on Coordinated and Multiple Views in Exploratory Visualization*, pp. 61–71. IEEE, 2007. doi: 10.1109/CMV.2007.20 3, 4
- [47] D. Sacha, M. Kraus, D. A. Keim, and M. Chen. VIS4ML: An ontology for visual analytics assisted machine learning. *IEEE Transactions on Visualization and Computer Graphics*, 25(1):385–395, 2019. doi: 10.1109/TVCG.2018.2864838 2
- [48] D. Sacha, A. Stoffel, F. Stoffel, B. C. Kwon, G. Ellis, and D. A. Keim. Knowledge generation model for visual analytics. *IEEE transactions on visualization and computer graphics*, 20(12):1604–1613, 2014. 3, 4
- [49] A. Satyanarayan, D. Moritz, K. Wongsuphasawat, and J. Heer. Vega-lite: A grammar of interactive graphics. *IEEE transactions on visualization and computer graphics*, 23(1):341–350, 2016. 3
- [50] H.-J. Schulz, T. Nocke, M. Heitzler, and H. Schumann. A design space of visualization tasks. *IEEE Transactions on Visualization and Computer Graphics*, 19(12):2366–2375, 2013. doi: 10.1109/TVCG.2013.120 3
- [51] M. Sedlmair, M. Meyer, and T. Munzner. Design study methodology: Reflections from the trenches and the stacks. *IEEE transactions on visualization and computer graphics*, 18(12):2431–2440, 2012. 1, 2
- [52] H. Stitz, S. Gratzl, H. Piringer, T. Zichner, and M. Streit. KnowledgePearls: Provenance-based visualization retrieval. *IEEE Transactions on Visualization and Computer Graphics*, 25(1):120–130, 2019. doi: 10.1109/TVCG.2018.2865024 2
- [53] L. Sun, L. He, S. Jia, Y. He, and C. You. Docagent: An agentic framework for multi-modal long-context document understanding. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 17701–17716, 2025. doi: 10.18653/v1/2025.emnlp-main.893 1, 2
- [54] J. Tang, Y. Zhou, T. Tang, D. Weng, B. Xie, L. Yu et al. A visualization approach for monitoring order processing in e-commerce warehouse. *IEEE Transactions on Visualization and Computer Graphics*, 28(1), 2022. doi: 10.1109/TVCG.2021.3114878 7
- [55] X. Wang, R. Huang, Z. Jin, T. Fang, and H. Qu. CommonsenseVIS: Visualizing and understanding commonsense reasoning capabilities of natural language models. *IEEE Transactions on Visualization and Computer Graphics*, 30(1), 2024. doi: 10.1109/TVCG.2023.3327153 7, 9
- [56] Y. Wu, D. Deng, X. Xie, M. He, J. Xu, H. Zhang et al. OBTracker: Visual analytics of off-ball movements in basketball. *IEEE Transactions on Visualization and Computer Graphics*, 29(1), 2023. doi: 10.1109/TVCG.2022.3209373 6
- [57] Y. Wu, S. Gao, S. Zhang, X. Dou, X. Wang, and Q. Li. From requirement to solution: Unveiling problem-driven design patterns in visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, 31(10):7089–7106, 2025. 1, 2, 3
- [58] Y. Xing, M. Ji, C. Dondi, and A. Abdul-Rahman. OwnershipTracker: A visual analytics approach to uncovering historical book ownership patterns. *IEEE Transactions on Visualization and Computer Graphics*, 32(1), 2026. doi: 10.1109/TVCG.2025.3634653 7
- [59] K. Xu, A. Ottley, C. Walchshofer, M. Streit, R. Chang, and J. Wenskovitch. Survey on the analysis of user interactions and visualization provenance. In *Computer Graphics Forum*, vol. 39, pp. 757–783. Wiley Online Library, 2020. 1, 2
- [60] K. Xu, Y. Wang, L. Yang, Y. Wang, B. Qiao, S. Qin et al. CloudDet: Interactive visual analysis of anomalous performances in cloud computing systems. *IEEE Transactions on Visualization and Computer Graphics*, 26(1), 2020. doi: 10.1109/TVCG.2019.2934613 7
- [61] R. Xu and Y. Yan. Agent skills for large language models: Architecture, acquisition, security, and the path forward. *arXiv preprint arXiv:2602.12430*, 2026. 1
- [62] L. Ying, A. Wu, H. Li, Z. Deng, J. Lan, J. Wu et al. Vaid: Indexing view designs in visual analytics system. In *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, pp. 1–15, 2024. 1, 2
- [63] J. Yuan, M. Liu, F. Tian, and S. Liu. Visual analysis of neural architecture spaces for summarizing design principles. *IEEE Transactions on Visualization and Computer Graphics*, 29(1), 2023. doi: 10.1109/TVCG.2022.3209404 6
- [64] X. Yue, J. Bai, Q. Liu, Y. Tang, A. Puri, K. Li et al. sPortfolio: Stratified visual analysis of stock portfolios. *IEEE Transactions on Visualization and Computer Graphics*, 26(1), 2020. doi: 10.1109/TVCG.2019.2934660 7
- [65] B. Zhang, K. Lazuka, and M. Murag. Equipping agents for the real world with Agent Skills. *Anthropic Engineering*, Oct. 2025. Published October 16, 2025. Accessed September 5, 2026. 2, 4
- [66] C. Zhang, X. Wang, C. Zhao, Y. Ren, T. Zhang, Z. Peng et al. PromotionLens: Inspecting promotion strategies of online e-commerce via visual analytics. *IEEE Transactions on Visualization and Computer Graphics*, 29(1), 2023. doi: 10.1109/TVCG.2022.3209440 7
- [67] J. Zhao, N. Cao, Z. Wen, Y. Song, Y.-R. Lin, and C. Collins. #FluxFlow: Visual analysis of anomalous information spreading on social media. *IEEE Transactions on Visualization and Computer Graphics*, 20(12):1773–1782, 2014. doi: 10.1109/TVCG.2014.2346922 4
- [68] Y. Zhao, X. Shu, L. Fan, L. Gao, Y. Zhang, and S. Chen. ProactiveVA: Proactive visual analytics with LLM-based UI agent. *IEEE Transactions on Visualization and Computer Graphics*, 32(1):451–461, 2026. doi: 10.1109/TVCG.2025.3642628 2
- [69] Y. Zhao, J. Wang, L. Xiang, X. Zhang, Z. Guo, C. Turkay et al. LightVA: Lightweight visual analytics with LLM agent-based task planning and execution. *IEEE Transactions on Visualization and Computer Graphics*, 31(9):6162–6177, 2025. doi: 10.1109/TVCG.2024.3496112 2
- [70] Y. Zhao, Y. Zhang, Y. Zhang, X. Zhao, J. Wang, Z. Shao et al. Leva: Using large language models to enhance visual analytics. *IEEE transactions on visualization and computer graphics*, 31(3):1830–1847, 2024. 2, 3